

UNIVERSITAT DE BARCELONA

MASTER THESIS

Borel Cardinality of Lascar Strong Types

Esperanza Buitrago Díaz

Written under supervision of professor
Enrique Casanovas

Submitted as a Master's thesis at

Department of Logic, History and Philosophy of Science
Universitat de Barcelona

June 2014

Acknowledgements

I would like to express my gratitude to my supervisor Enrique Casanovas for his useful comments, remarks and engagement through the learning process of this master thesis. Furthermore I would like to thank Rafel Farré for introducing me to the topic as well for the support on the way. Also, I like to thank all my classmates, professors and members of the Model Theory seminar, who have willingly shared their time and knowledge during the process of building concepts that were completely new to me.

Finally, I would like to thank my loved ones: my mother and father, my sisters and niece, Arf, Juni, Paola and Roberto, Viviana, Yerko and Vicente, all those who have supported me throughout the entire process, both by keeping me harmonious and helping me putting pieces together.

Contents

Acknowledgements	ii
Contents	iv
1 Preliminaries	3
1.1 The Lascar strong type	4
1.2 Topology	6
1.2.1 Action of Polish groups	11
2 The Lascar strong type in the Stone space	15
3 Small Case	19
4 The countable case	22

Introduction

A relation R is *bounded* if for some cardinal κ there is no sequence $(a_i : i < \kappa)$ such that $\neg Ra_i a_j$, for all $i < j < \kappa$. A *strong type* is a class of a bounded equivalence relation on tuples of the monster model $\mathfrak{C}$ of a complete theory T . A *Kim-Pillay strong type* is a class of the least bounded type-definable (over $\emptyset$) equivalence relation $\equiv^{\text{kp}}$. This type may be characterized as the finest notion of strong type for which the corresponding quotient is a compact Hausdorff space when it is equipped with the so-called logic topology [11]. A *Lascar strong type* is simply a class of the smallest equivalence relation $\equiv^{\text{ls}}$ which is bounded and invariant under all automorphisms of the monster model.

It took some years to find examples of theories for which Kim-Pillay and Lascar strong type do not coincide, despite the fact that Lascar strong type does not commonly come equipped with a Hausdorff topology as Kim-Pillay strong type does. Thereafter, it was suggested that Lascar strong types could be described from descriptive set-theoretic point of view: using properties of Borel equivalence relations defined on Polish spaces or, more generally, Strong Choquet games.

Furthermore, Krupinsky, Pillay and Solecki conjectured in [16] that for every tuple a in the monster model, if its Lascar strong type $[a]_{\text{ls}}$ is not the same as its Kim-Pillay strong type $[a]_{\text{kp}}$, then the restriction of $\equiv^{\text{ls}}$ to $[a]_{\text{kp}}$ is nonsmooth, this implies that whenever Kim-Pillay strong type is different to Lascar strong type by at least one element, then they differ in at least $2^{\aleph_0}$ elements.

This master's thesis is meant to serve, first of all, as an introduction to the aforementioned problem. Not just in the sense of setting up the necessary definitions and properties from Model Theory in order to understand the problem, but also some basic notions and results from Descriptive Set Theory that connect Lascar and Kim-Pillay equivalence relations over $\mathfrak{C}$ to associated Borel equivalence relations over some Polish space, respectively. Even though Kaplan, Miller and Simon in [11] solved such question for theories with languages of any cardinality, we are going to focus on those with countable languages.

Chapter 1

Preliminaries

Let T be a complete first order theory over the language $\mathcal{L}$. The monster model $\mathfrak{C}$ of T will always be considered as a proper class and models of T will always be considered as sets.

A formula $\varphi = \varphi(x)$ is said to be *over* A if it has parameters in A , in such case we write $\varphi \in \mathcal{L}(A)$. If $A = \emptyset$, we say that the formula is without parameters and we will simply write $\varphi \in \mathcal{L}$. We will usually understand that φ has no parameters and x is an α -tuple, for some ordinal α , unless the contrary is stated. For a formula $\varphi \in \mathcal{L}(\mathfrak{C})$ we write $\models \varphi(a)$ or $a \models \varphi$ to mean that $\mathfrak{C} \models \varphi(a)$. A *partial type* $\pi = \pi(x)$ is a consistent set of formulas, and a partial type $p = p(x)$ over A is *complete* if it contains φ or $\neg\varphi$, for any formula $\varphi = \varphi(x)$ over A . For any two partial types p, q we denote with $p \cup q$ to the set $p \cup q \cup \{\varphi \wedge \psi : \varphi \in p, \psi \in q\}$. Observe that $p \cup q$ is not necessarily a partial type.

Given an α -tuple of variables x , the Stone space of all complete types over A is denoted with $S_\alpha(A)$, or simply, $S(A)$, whenever α is not ambiguous from the context. Sometimes we want to make clear when x is a finite tuple, i.e. $\alpha = n$, for some $n \in \mathbb{N}$, in such case we denote the Stone space with $S_n(A)$. Since the subsets $[\varphi] = \{p \in S(A) : \varphi \in p\}$ of $S(A)$, for a formula φ over A , form a basis of clopen sets, $S(A)$ is, in general, a compact Hausdorff totally disconnected topological space. For any partial type Σ over A given, the set $[\Sigma] = \{p \in S(A) : \Sigma \subseteq p\}$ is a closed subset of $S(A)$.

The group of automorphisms of a model M (which could be $\mathfrak{C}$) is denoted with $\text{Aut}(M)$, and the subgroup of automorphisms of a model M that fix pointwise a set $A \subseteq M$ is denoted with $\text{Aut}(M/A)$. The image of an object by some automorphism $\sigma \in \text{Aut}(\mathfrak{C}/A)$ is called its *A-conjugate*. Thus, if $\varphi(x, y) \in \mathcal{L}$, the *A-conjugate* of $\varphi(x, a)$ by σ is the formula $\varphi(x, a)^\sigma = \varphi(x, \sigma(a))$. This defines, in fact, a natural action on the space of

types associated to $\text{Aut}(\mathfrak{C}/A)$. For any $a, b \in \mathfrak{C}^\alpha$, we write $a \equiv_A b$ to express equality of types over A . Recall that $a \equiv_A b$ if and only if there is some $\sigma \in \text{Aut}(\mathfrak{C}/A)$ such that $\sigma(a) = b$.

For a linearly ordered set $(I, <)$, a sequence $(a_i : i \in I)$ of tuples of the same length is *A-indiscernible* if for every natural number n and every $i_0, \dots, i_n, j_0, \dots, j_n \in I$ such that $i_0 < \dots < i_n$ and $j_0 < \dots < j_n$, we have that $a_{i_0} \dots a_{i_n} \equiv_A a_{j_0} \dots a_{j_n}$. If $A = \emptyset$ we simply say $(a_i : i \in I)$ is an indiscernible sequence. The existence of *A*-indiscernible sequences is proved using Ramsey's theorem (see p.64, [24]).

1.1 The Lascar strong type

Let α be an ordinal. As said in before, a relation R on $\mathfrak{C}^\alpha$ is said to be *bounded* if there exists a cardinal κ such that there is no sequence $(a_i : i < \kappa)$ of α -tuples for which $\neg R(a_i, a_j)$ for all $i < j < \kappa$. It is easy to check that any intersection of bounded relations is bounded. If κ is a natural number, R is called *finite*. In particular, an equivalence relation is bounded if it has no more than κ -many equivalence classes, for some cardinal κ . R is said to be *0-definable*, if there is a formula $\varphi(x, y) \in \mathcal{L}$, for α -tuples of variables x and y , such that for all $a, b \in \mathfrak{C}^\alpha$ we have that Rab if and only if $\models \varphi(a, b)$. One can check that for 0-definable relations, finiteness coincides with boundedness. R is called *0-type-definable* if there exists a partial type $\Gamma(x, y)$ of formulas in $\mathcal{L}$, for α -tuples x and y , such that for all $a, b \in \mathfrak{C}^\alpha$ we have that Rab if and only if $\models \Gamma(a, b)$. Finally, R is said to be *A-invariant* if it is preserved under any $\sigma \in \text{Aut}(\mathfrak{C}/A)$, that is, if Rab then $R\sigma(a)\sigma(b)$. There are at most $2^{2^{|T|+|A|+\alpha}}$ *A*-invariant relations on α -tuples (Lemma 9.3, [3]).

The *Kim-Pillay equivalence relation* $\equiv_{\alpha}^{\text{kp}}$ on $\mathfrak{C}^\alpha$ is the intersection of all type-definable (over $\emptyset$) equivalence relations on $\mathfrak{C}^\alpha$. We will omit α whenever it is clear from the context. A *Kim-Pillay strong type* is an equivalence class of $\equiv_{\alpha}^{\text{kp}}$, thus, $[a]_{\text{kp}}$ denotes the Kim-Pillay strong type of a , for $a \in \mathfrak{C}^\alpha$. The *Lascar equivalence relation* $\equiv_{\alpha}^{\text{ls}}$ on $\mathfrak{C}^\alpha$ is the intersection of all A -invariant and bounded equivalence relations on $\mathfrak{C}^\alpha$. We will omit α whenever it is clear from the context. A *Lascar strong type* is an equivalence class of the Lascar equivalence relation and the Lascar strong type of $a \in \mathfrak{C}^\alpha$ is denoted with $[a]_{\text{ls}}$.

Since every equivalence relation that is type-definable (over $\emptyset$) is A -invariant, we have that $[a]_{\text{ls}} \subseteq [a]_{\text{kp}}$. The relation of having the same Lascar strong type coincides with the transitive clousure of the relation of starting indiscernible sequences (Proposition 9.9, [3]). Moreover, if we denote with $\text{Autf}(\mathfrak{C})$ the subgroup of $\text{Aut}(\mathfrak{C})$ generated by

automorphisms fixing pointwise models of T , then a and b have the same Lascar strong type if and only if there exists $\sigma \in \text{Aut}(\mathfrak{C})$ that maps a to b (Corollary 9.15, [3]). An element of $\text{Aut}(\mathfrak{C})$ is called a *strong automorphism*. If we consider the set $nc(x, y)$ of all formulas that define a finite and symmetric relation in α -tuples of variables x, y , then the Lascar equivalence relation is the transitive closure of the relation that is defined by $nc(x, y)$. In other words, the Lascar equivalence relation could be defined by the infinite disjunction $\bigvee_{n < \omega} nc^n(x, y)$, where $nc^n(x, y)$ is the n -times composition of $nc(x, y)$ (Lemma 9.7, [3]). Finally, for tuples a, b of the same length, put $d(a, b) = 0$ if $a = b$; $d(a, b) = n$ if $a \neq b$ and n is the least number for which there are infinite indiscernible sequences $I_1, \dots, I_n$ and tuples $c_0 = a, c_1, \dots, c_n = b$ such that $c_{i-1}, c_i \in I_i$; otherwise, put $d(a, b) = \infty$. Clearly, $d(a, b) \leq n$ if and only if $\models nc^n(a, b)$; so, $a \equiv_{\text{ls}} b$ if and only if $d(a, b) < \infty$.

Keren showed that for any strong automorphism σ , there is some $m \in \mathbb{N}$ such that for every tuple c (of any length), $d(c, \sigma(c)) \leq m$. In this case we say that m *bounds* σ , (Section 3.1, [13]). The proof of this fact is contained in the following remark.

Remark 1.1. For $a, b \in \mathfrak{C}^\alpha$ and $d(a, b) \leq n$, there is a strong automorphism σ bounded by $2n$ such that $\sigma(a) = b$.

Proof. First, assume that $d(a, b) \leq 1$. Then $\models nc(a, b)$, and hence there is a model M for which $a \equiv_M b$. Let $\sigma \in \text{Aut}(\mathfrak{C}/M)$ be such that $\sigma(a) = b$. Hence, σ is a strong automorphism bounded by 2. Secondly, suppose that $n > 1$. There are $c_0 = a, c_1, \dots, c_{n-1}, c_n = b$ tuples of the same length such that $d(c_i, c_{i+1}) \leq 1$. Since for each pair c_i, c_{i+1} there is a strong automorphism σ_{i+1} bounded by 2 mapping c_i to c_{i+1} , the strong automorphism $\sigma = \sigma_n \dots \sigma_1$ is bounded by $2n$. $\square$

Lascar and Pillay introduced the notion of G -compactness for theories (notion that is studied by Peláez in his Ph.D. thesis [20]). It is known that G -compactness is equivalent to the fact that $\equiv_{\text{ls}}$ coincides with $\equiv_{\text{kp}}$. In consequence of the work by Casanovas, Lascar, Pillay and Ziegler in [4] that showed that there are theories which are not G -compact, we have an example of a theory in which the Lascar strong type is not equal to the Kim-Pillay strong type. Furthermore, Newelski proved in [19] that equality of Lascar strong types is type-definable if and only if for some $n < \omega$ it is defined by $nc^n(x, y)$, (Corollary 9.23, [3]). In addition, Newelski showed that for small theories (i.e. theories such that $|S_k(\emptyset)| \leq \aleph_0$, for all $k \in \mathbb{N}$), for every natural number n and $a \in \mathfrak{C}^n$ we have that $[a]_{\text{ls}} = [a]_{\text{kp}}$.

1.2 Topology

For this section we will use mainly [12] and [23] as references to explain certain basic notions and results from Descriptive Set Theory.

Let (X, τ) be a topological space. X is *metrizable* if there is a metric d on X so that τ is the topology of (X, d) , in this case we say that the metric is *compatible* with τ . If a metric d is compatible with τ , the metric d' defined as $\frac{d}{1+d}$ is so that $d' \leq 1$ and is compatible with τ . X is *second-countable* if it has a countable base. A subset D of X is *dense* if it meets every nonempty open set. If X contains a countable dense set, we say that X is *separable*. Every second-countable space is separable (in general the converse does not hold, unless X is metrizable). X is T_1 if every singleton is closed. It is *Hausdorff* if every two points have neighborhoods separating them (i.e. there are two disjoint open sets containing each point). X is *Regular* if for every closed subset F and every point $x \notin F$, there exist two disjoint open sets such that x belongs to one and F is contained in the other. X is *normal* if every two disjoint closed sets can be separated by two disjoint open sets. Finally, X is *compact* if every covering of X by open subsets has a finite subcovering of X .

Y is a G_δ set in X if it is a countable intersection of open subsets of X ; Y is an F_σ set in X if its relative complement with respect to X is a G_δ , or equivalently, if it is a countable union of closed subsets of X . Recall that closed subsets of a compact space are compact sets. A continuous image of a compact space is compact. In particular, if $f : X \rightarrow Y$ is continuous mapping from a compact space to a Hausdorff space, then f preserves closed and F_σ sets.

In many books of basic topology, for instance in [18], one can find the proof of the following important metrization theorem.

Theorem 1.2 (Urysohn's theorem). *Let X be a second countable topological space. Then, X is metrizable if and only if X is T_1 and regular.*

Definition 1.3. A topological space (X, τ) is *completely metrizable* if there is a metric d compatible with τ such that every Cauchy sequence converges. X is a *Polish space* if it is a separable completely metrizable space.

Important examples of Polish spaces are $\mathbb{R}$, $\mathbb{C}$, $\mathbb{I} = [0, 1]$ and any compact metrizable space. Even countable sets with the discrete metric such as $2 = \{0, 1\}$ or $\mathbb{N}$ are Polish spaces. As result, subspaces of a Polish space (with the induced topology) and countable products (with the product topology) of Polish spaces are Polish spaces. Thus, the

Hilbert cube, the *Cantor space* and the *Baire space* defined by $\mathcal{H} = \mathbb{I}^\omega$, $\mathcal{C} = 2^\omega$ and $\mathcal{B} = \mathbb{N}^\omega$, respectively, are Polish spaces.

There are many classical results about Polish spaces in relation with $\mathcal{H}$, $\mathcal{C}$ or $\mathcal{B}$. We will sketch the proof of one of those in Theorem 1.7, because the Cantor Space will play a fundamental role here. It is convenient to mention that for some natural number k and a finite sequence $r \in 2^k$, the sets

$$O_r = \{s \in 2^\omega : s \supseteq r\},$$

form a basis for the Cantor space.

Remark 1.4. Several other metrization theorems follow as simple corollaries from the Urysohn's theorem. Namely, a compact Hausdorff space is metrizable if and only if it is second-countable. We know that the Stone space is a compact Hausdorff space, but it may not always be second countable. Of course, if T is countable and we consider a model M and an ordinal α that are countable, the Stone space $S_\alpha(M)$ is second countable and hence metrizable. In addition to that, a metric space is compact if and only if it is complete and totally bounded (Theorem 45.1, [18]). Consequently, if T , M and α are countable, the Stone space $S_\alpha(M)$ is a Polish space.

Theorem 1.5. (Theorem 3.11, [12]) *If X is metrizable and $Y \subseteq X$ is completely metrizable, then Y is a G_δ set in X . Conversely, if X is completely metrizable and $Y \subseteq X$ is a G_δ set, then Y is completely metrizable.*

In particular, a subspace of a Polish space is Polish if and only if it is a G_δ subset.

Let's first recall Kuratowski's Theorem.

Theorem 1.6. (Theorem 3.8, [12]) *Let X be a metrizable, Y be completely metrizable, $A \subseteq X$, and $f : A \rightarrow Y$ be continuous. Then there is a G_δ subset G of X with $A \subseteq G \subseteq \bar{A}$ and a continuous extension $g : G \rightarrow Y$ of f .*

(Sketch of the proof of Theorem 1.5). First assume that Y is completely metrizable and consider the continuous mapping $id_Y : Y \rightarrow Y$. By Kuratowski's theorem there is a G_δ subset G of X and a continuous extension $g : G \rightarrow Y$ of id_Y . Since Y is completely metrizable, every Cauchy sequence $(y_n)_{n \in \mathbb{N}}$ of Y converges and so its limit belongs to G . So, Y is dense in G . This implies that $g = id_Y$ and $G = Y$.

Secondly, let $Y = \bigcap_{n < \omega} U_n$, where each U_n is open in X . Consider $F_n = X \setminus U_n$ and let d be a compatible metric for X . Denote with $d(x, F_n) = \inf\{d(x, y) : y \in F_n\}$ and notice that it is nonzero for each $n \in \mathbb{N}$ and point $x \in Y$. Thus,

$$d'(x, y) = d(x, y) + \sum_{n=0}^{\infty} \min \left\{ \frac{1}{2^{n+1}}, \left| \frac{1}{d(x, F_n)} - \frac{1}{d(y, F_n)} \right| \right\}$$

defines a complete metric compatible with Y . $\square$

Theorem 1.7. (Theorem 4.18, [12]) *Every nonempty compact metrizable space is a continuous image of the Cantor space $\mathcal{C} = 2^\omega$.*

(Sketch of proof). The function $f(x) = \sum_{n \in \mathbb{N}} \frac{x(n)}{2^{n+1}}$ maps $\mathcal{C}$ continuously onto $\mathbb{I}$. So, $(x_n) \mapsto (f(x_n))$ is a continuous mapping from $\mathcal{C}^\omega$ to $\mathcal{H} = \mathbb{I}^\omega$. Since $\mathcal{C}$ is homeomorphic to $\mathcal{C}^\omega$, $\mathcal{H}$ is a continuous image of $\mathcal{C}$. Since every compact metrizable space is homeomorphic to a compact subset of $\mathcal{H}$, for every compact metrizable space X there is a closed subset F of $\mathcal{C}$ and a continuous surjection of F onto X . But F is a retract of $\mathcal{C}$, this means that there is a continuous surjection of $\mathcal{C}$ onto F such that $x \mapsto x$, for every $x \in F$. Hence, X is a continuous image of $\mathcal{C}$. $\square$

Let (X, τ) be a topological space. A subset Y of X is called *nowhere dense* if its closure $\bar{Y}$ has empty interior (equivalently, if $X \setminus \bar{Y}$ is dense). Y is *meager* if it is a countable union of nowhere dense sets. The complement of a meager set is called *comeager*. The class of *Borel sets* of X is the σ -algebra generated by the open subsets of X . The class of Borel sets is countably generated when X is second countable and it contains all open, closed, G_δ and F_σ subsets of X . An α -ary relation R on a topological space X is *Borel* if it is a Borel subset of X^α (as topological space). For topological spaces X and Y , a function $f : X \rightarrow Y$ is *Borel (measurable)* if the inverse image of a Borel (equivalently: open or closed) set is Borel. Notice that if f is a Borel function, then its graph is a Borel subset of the product topology $X \times Y$. If Y has a countable basis $\{V_n : n \in \omega\}$, it is enough to require that $f^{-1}[V_n]$ is Borel, for each $n \in \omega$.

Definition 1.8. Let X and Y be Polish spaces. Let E_0, E_1 and F_0, F_1 be binary relations of α -tuples of X and Y , respectively. We say that a function $f : X \rightarrow Y$ is a *reduction* of (E_0, E_1) to (F_0, F_1) if

$$\langle x_0, x_1 \rangle \in E_i \quad \text{implies that} \quad \langle f(x_0), f(x_1) \rangle \in F_i, \quad \text{for } i = 0, 1.$$

In particular, for E and F relations on α -tuples of X and Y , respectively, we say that f is a *reduction of E to F* if f is a reduction of $(E, \sim E)$ to $(F, \sim F)$ (where $\sim E$ represents $X^\alpha \times X^\alpha \setminus E$).

Let E and F be Borel equivalence relations. We say that E is *Borel reducible* to F if there is a Borel reduction $f : X \rightarrow Y$ of E to F . In such case, we denote

$$E \leq_B F$$

E is *continuously reducible* to F if f is a continuous Borel reduction of E to F . In this case we denote

$$E \sqsubseteq_B F$$

We say that E and F are *Borel bireducible*, denoted by $E \sim_B F$ when $E \leq_B F$ and $F \leq_B E$. Finally, we denote with $E <_B F$ if $E \leq_B F$ and $E \not\sim_B F$.

Remark 1.9. Let x, y be points of a Polish space X and U, V be two disjoint neighborhoods of x and y , respectively. Clearly, $U \times V$ is a neighborhood of $\langle x, y \rangle$. So, if $\Delta(X) = \{(x, x) : x \in X\}$ then $\sim \Delta(X)$ is open and hence $\Delta(X)$ is closed. Moreover, if we consider the sets $1, 2, \dots, \omega$ with the discrete topology and 2^ω with the product topology, we obtain that

$$\Delta(1) <_B \Delta(2) <_B \dots <_B \Delta(\omega) <_B \Delta(2^\omega).$$

It is clear that $\Delta(1) \leq_B \Delta(2) \leq_B \dots \leq_B \Delta(\omega) \leq_B \Delta(2^\omega)$ because the inclusion function is continuous, and so it is Borel. Now, let X, Y be $1, 2, \dots, \omega$ or 2^ω be such that $|X| > |Y|$. If $\Delta(X) \leq_B \Delta(Y)$ by a Borel reduction f , then there are two distinct elements x_0, x_1 in X for which $f(x_0) = f(x_1)$. Since $x_0 \neq x_1$, by Borel reducibility, $f(x_0) \neq f(x_1)$ which is a contradiction. Hence, $\Delta(X) \not\sim_B \Delta(Y)$.

Definition 1.10. A Borel equivalence relation E on a Polish space is *smooth* whenever $E \leq_B \Delta(2^\omega)$.

Proposition 1.11. E is smooth if and only if there are separating Borel sets, i.e. a countable family of Borel sets $\{B_i : i < \omega\}$ such that

$$xEy \text{ iff } x \in B_i \leftrightarrow y \in B_i, \text{ for all } i < \omega$$

Proof. From left to right. Assume that E is smooth and let $f : X \rightarrow \mathcal{C}$ be a Borel function that reduces E to $\Delta(2^\omega)$. For $i < \omega$, the set $\{s \in 2^\omega : s(i) = 1\}$ is Borel in $\mathcal{C}$ and let B_i be its inverse image by f . For $x, y \in X$, xEy if and only if $f(x)(i) = f(y)(i)$, for all $i < \omega$ which is equivalent to say that $x \in B_i$ iff $y \in B_i$.

From right to left, it is enough to check that the inverse image by f of the basic open sets $O_r = \{s \in 2^\omega : s \supseteq r\}$, where $r \in 2^k$, for some natural number k , is a Borel subset of X . For $i < k$, define $X_i = \{x \in X \mid x \notin B_i \text{ iff } r(i) = 0\} \cup \{x \in X \mid x \in B_i \text{ iff } r(i) = 1\}$. Observe that X_i is Borel, and hence $f^{-1}[O_r] = \bigcap_{i < k} X_i$. Therefore, f is Borel. Thus, xEy if and only if for all $i < \omega$, $x \in B_i$ iff $y \in B_i$, by definition of f , this is equivalent to say that $f(x)(i) = f(y)(i)$, for all $i < \omega$. $\square$

Proposition 1.12. (Silver dichotomy, [22]) For all Borel equivalence relations E , either $E \leq_B \Delta(\omega)$ or $\Delta(2^\omega) \sqsubseteq_B E$. It follows that $\Delta(2^\omega)$ is the successor of $\Delta(\omega)$.

Let X be a set. We denote the projection of $A \subseteq X \times X$ to X by

$$\exists^Y A = \{x \in X : \exists y \langle x, y \rangle \in A\}.$$

A subset of a Polish space X is called *analytic* if it is the projection of a Borel subset of $X \times X$. For an equivalence relation E on X and a Borel subset B of X , the E -saturation of B is the set defined as follows:

$$B^E := \{x \in X : \exists y (y \in B \wedge xEy)\}.$$

Observe that if E is a Borel equivalence relation on X we have that a point x belongs to B^E if and only if there is some $y \in X$ such that $y \in B$ and xEy , this means that there is $y \in X$ for which $\langle x, y \rangle \in B \times B$ and xEy . Thus, the E -saturation of a Borel set B is the projection of the Borel set $B^2 \cap E$, and hence it is analytic.

The separation theorems are among the most important results on analytic sets. Here we recall what is known as the First Separation Theorem for analytic sets:

Theorem 1.13 (Theorem 4.4.1, [23]). *Let A and B be disjoint analytic subsets of a Polish space X . Then, there is a Borel subset C of X such that $A \subseteq C$ and $B \cap C = \emptyset$.*

Proposition 1.14. *Let E be an equivalence relation on a Polish space. If E is closed then E is smooth.*

Proof. For $i < \omega$, let U_i, V_i be open subsets of X such that $X \setminus E = \bigcup_{i < \omega} U_i \times V_i$. Let U_i^E and V_i^E be the E -saturation of U_i and V_i , respectively. These sets are analytic. Furthermore, if $x \in U_i^E \cap V_i^E$, there are $y \in U_i$ and $y' \in V_i$ such that xEy and xEy' ; this implies that yEy' , but $\langle y, y' \rangle \in U_i \times V_i$ which is a contradiction. So, $U_i^E \cap V_i^E = \emptyset$.

For $i < \omega$, there is a Borel set $U_i^0 \supseteq U_i^E$ such that $U_i^0 \cap V_i^E = \emptyset$, by Theorem 1.13. Recursively, assume that for $i, j < \omega$ we have constructed U_i^j and let U_i^{j+1} be a Borel subset that contains the E -saturation of U_i^j and it is disjoint from V_i^E . So, the set $B_i = \bigcup_{j < \omega} U_i^j$ is Borel.

We claim that xEy if and only if $x \in B_i$ iff $y \in B_i$, for all $i < \omega$. For one direction, assume that $\langle x, y \rangle \notin E$, there is $i < \omega$ such that $x \in U_i$ and $y \in V_i$, so, $x \in B_i$ and $y \notin B_i$. To check the other direction, assume that xEy . For $i < \omega$, if $x \in B_i$, then there is some $j < \omega$ such that $x \in U_i^j$. By construction, B_i contains the E -saturation of U_i^j , and hence $y \in B_i$. $\square$

Denote with $\mathbb{E}_0$ the eventually equality relation on $\mathcal{C}$ defined as follows:

$$s \mathbb{E}_0 t \quad \text{iff} \quad \exists n < \omega \forall m > n \quad s(m) = t(m)$$

Let k be a natural number, $r \in 2^k$ and $s \in 2^\omega$. The sequence

$$t(i) = \begin{cases} r(i) & i < k \\ s(i) & \text{otherwise} \end{cases}$$

clearly belongs to $[s]_{\mathbb{E}_0} \cap O_r$, where $[s]_{\mathbb{E}_0}$ denotes the $\mathbb{E}_0$ -class of s . Hence, every $\mathbb{E}_0$ -class is dense in $\mathcal{C}$.

Proposition 1.15. $\mathbb{E}_0$ is nonsmooth.

Sketch of the proof. Assume that $\mathbb{E}_0$ is smooth. Let $\{B_i : i < \omega\}$ be a separating family of Borel subsets, according with Corollary 1.11. For $i < \omega$, if B_i were nonmeager, then there is a natural number k and a finite sequence $r \in 2^k$ such that $O_r \setminus B_i$ is meager, since every Borel set has the Baire property. For any $r' \in 2^k$, there is an homeomorphism of $\mathcal{C}$ taking O_r to $O_{r'}$ fixing all the $\mathbb{E}_0$ -equivalence classes. Since every B_i is closed under $\mathbb{E}_0$ -equivalence classes, $O_{r'} \setminus B_i$ is meager. If $s \notin B_i$, then $s \in O_{s|k}$, and so $s \in \bigcup_{r \in 2^k} O_r \setminus B_i$. Thus, $2^\omega \setminus B_i = \bigcup_{r \in 2^k} O_r \setminus B_i$, and hence B_i is comeager. Therefore, every B_i is either meager or comeager. Thus, the set

$$\bigcap \{B_i : i < \omega, B_i \text{ is comeager}\} \cap \bigcap \{2^\omega \setminus B_i : i < \omega, B_i \text{ is meager}\}$$

is a comeager $\mathbb{E}_0$ -class. This implies a contradiction. $\square$

Furthermore, up to Borel bireducibility, the eventually equality relation $\mathbb{E}_0$ is the only nonsmooth Borel equivalence relation which is hyperfinite (this means that it can be represented as the increasing union of a sequence of Borel equivalence relations with finite equivalence classes, or equivalently, it is induced by the orbits of a single Borel automorphism). The proof of this fact may be found in [6].

Proposition 1.16. (Harrington-Kechris-Louveau dichotomy, [8]) For every Borel equivalence relation E on a Polish space, either $E \leq_B \Delta(2^\omega)$ or $\mathbb{E}_0 \subseteq_B E$.

Corollary 1.17. If E is a Borel equivalence relation on a Polish space such that all its classes are G_δ sets, then E is smooth.

Proof. Assume that E is nonsmooth. Let $f : 2^\omega \rightarrow X$ be a continuous map that reduces $\mathbb{E}_0$ to E , given by Proposition 1.16. Since f is Borel, for $x \in X$ we have that the inverse image of $[x]_E$ must be a G_δ subset of 2^ω , hence it contains a dense subset of 2^ω , and so it is comeager. This is a contradiction because the complement of $f^{-1}([x]_{\mathbb{E}_0})$ contains an $\mathbb{E}_0$ -class. $\square$

1.2.1 Action of Polish groups

Let Γ be a group and X be a Polish space. An *action of Γ on X* is a continuous map $\otimes : \Gamma \times X \rightarrow X$ such that $1_\gamma \cdot x = x$ and $\sigma(\tau \cdot x) = \sigma\tau \cdot x$, for any $x \in X$ and $\sigma, \tau \in \Gamma$.

A pair $(X, \circledast)$ is called *topological Γ -space*, and it has an associated equivalence relation E_Γ^X defined as follows:

$$xE_\Gamma^X y \quad \text{iff} \quad \sigma \cdot x = y, \text{ for some } \sigma \in \Gamma$$

For $x \in X$, the set $\Gamma \cdot x := \{\sigma \cdot x : \sigma \in \Gamma\}$ is an *orbit of x* (associated with the action of Γ on X) and denotes the E_Γ^X -equivalence class of x . Notice that it defines a subspace of X , and so it is a Polish space. Denote with $X/\circledast \Gamma$, or simply, X/Γ , the set of all orbits of the action Γ .

Given a sequence $\langle \gamma_i : i < n \rangle$ of elements of Γ , for some natural number n , we write

$$\prod_{i < n} \gamma_i = \begin{cases} 1_\Gamma & \text{if } n = 0 \\ \gamma_0 \dots \gamma_{n-1} & \text{otherwise} \end{cases}$$

Theorem 1.18. *Let X be a perfect Polish space (i.e. X has no isolated points), Γ be a group of homeomorphisms of X with a dense orbit, and R be a meager subset of X^2 . Then, there is a continuous injective homomorphism $f : \mathcal{C} \rightarrow X$ from $(\mathbb{E}_0, \sim \mathbb{E}_0)$ to $(E_\Gamma^X, \sim R)$.*

Proof. Let $\{A_n : n < \omega\}$ be a family of nowhere dense subsets of X^2 such that R is $\bigcup_{n < \omega} A_n$. As X has no isolated points, singletons of X^2 are nowhere dense sets. Particularly, $\Delta(X)$ is a nowhere dense set. Observe that the set $F_n = \bar{A}_n \cup \bar{A}_n^{-1} \cup \Delta(X)$ is closed and nowhere dense subset of X^2 , for each $n < \omega$. Thus, without loss of generality, there is an increasing sequence $(F_n : n \in \omega)$ of closed nowhere dense subsets of X^2 such that $\bigcup_{n < \omega} F_n$ contains R . In other words, taking $U_n = X^2 \setminus F_n$, there is a decreasing sequence $(U_n : n \in \omega)$ of comeager, irreflexive and symmetric subsets of X^2 such that $\bigcap_{n < \omega} U_n$ is disjoint from R .

We want to construct a sequence of homeomorphisms $\gamma_n \in \Gamma$ with an associated element $\gamma_s = \prod_{i < n} \gamma_i^{s(i)}$, for $s \in 2^n$, along with points $x_n \in X$ and open neighborhoods V_n of x_n such that:

1. $\bar{V}_{n+1} \subseteq V_n \cap (\gamma_n^{-1} \cdot V_n)$,
2. $\text{diam}(\gamma_s \cdot V_{n+1}) < \frac{1}{n}$, for each $s \in 2^n$,
3. for each $s, t \in 2^{n+1}$, if $s(n) \neq t(n)$ then $(\gamma_s \cdot V_{n+1}) \times (\gamma_t \cdot V_{n+1}) \subseteq U_n$.

Construction. Fix $x_0 \in X$ and let $V_0 = X$. Suppose that for n we have already found x_n its open neighborhood V_n and γ_i , for $i < n$, according to conditions 1-3.

Consider the set

$$W_n = \bigcap \{(\gamma_s \times \gamma_t)^{-1} \cdot U_n : (s, t) \in 2^n \times 2^n\}$$

Using continuity of the elements of Γ and the fact that U_n is dense open, each set of the form $(\gamma_s \times \gamma_t) \cdot U_n$ is dense open and so is W_n .

Let $\mathcal{O}$ be a dense orbit of Γ . There is a pair $\langle x, y \rangle$ in $W_n \cap V_n^2 \cap \mathcal{O}^2$. Hence, there are some γ_n in Γ and $x_{n+1} \in V_n$ such that $y = \gamma_n \cdot x_{n+1}$. That means, $\langle x_{n+1}, \gamma_n \cdot x_{n+1} \rangle \in V_n$, or equivalently, $x_{n+1} \in V_n \cap \gamma_n^{-1} V_n$.

Considering that $\langle x_{n+1}, \gamma_n \cdot x_{n+1} \rangle \in W_n$, then $\langle \gamma_s \cdot x_{n+1}, \gamma_t \gamma_n \cdot x_{n+1} \rangle \in U_n$. Consequently, in order to define V_{n+1} we can take the intersection of neighborhoods of x_{n+1} satisfying conditions 1-3. $\square$

We claim now that the function f defined for every $s \in 2^\omega$ as $f(s) = \lim_{n \rightarrow \infty} \gamma_{s \upharpoonright n} \cdot x_n$ works. Thus, we have to check the following:

- f is an homomorphism, because Γ is an action on X .
- Let $s \in 2^\omega$ and W be an open neighborhood of $f(s)$. As $f(s) = \lim_{n \rightarrow \infty} \gamma_{s \upharpoonright n} \cdot x_n$, there some N such that $\gamma_{s \upharpoonright n} \cdot x_n \in W$, for all $n > N$. In other words, $\gamma_{s \upharpoonright n} \cdot V_n \cap W$ is nonempty, for every $n > N$. Chose N in a way such that $\gamma_{s \upharpoonright N} \cdot V_N \subseteq W$, using condition 2. So, for any $t \in 2^\omega$ extending $s \upharpoonright N$ we have that $\gamma_{t \upharpoonright n} \cdot x_n \in \gamma_{t \upharpoonright n} \cdot V_n$ and $\bar{V}_n \subseteq V_N$, for all $n > N$, and hence $\gamma_{t \upharpoonright n} \cdot V_n \subseteq \gamma_{s \upharpoonright N} \cdot V_N$. That is, $f(t) \in W$. We have proved that the direct image of the basic open $O_{s \upharpoonright N}$ is a subset of W . Therefore, f is continuous.
- f is injective: 1, 2 and completeness of X assure us that $f(s) \in \gamma_{s \upharpoonright n} \cdot X_{n+1}$. If $s \neq t$, let n be the minimum natural number for which $s(n) \neq t(n)$. According with condition 3, we have that $\gamma_{s \upharpoonright n+1} \cdot X_{n+1} \times \gamma_{t \upharpoonright n+1} \cdot X_{n+1} \subseteq U_n$. By condition 1, $\langle f(s), f(t) \rangle$ is an element of $\gamma_{s \upharpoonright n+1} \cdot X_{n+1} \times \gamma_{t \upharpoonright n+1} \cdot X_{n+1}$, which is contained by U_n . Since U_n is irreflexive, $f(s) \neq f(t)$.
- If $\langle s, t \rangle \in \mathbb{E}_0$ then $\langle f(s), f(t) \rangle \in E_\Gamma^X$: let $r \in 2^\omega$ be such that $s = s' \frown r$ and $t = t' \frown r$ for some natural number k for which $s', t' \in 2^k$. First, observe that $\gamma_s = \prod_{i < n} \gamma_i^{s' \frown r(i)} = \prod_{i < k} \gamma_i^{s'(i)} \prod_{i < n} \gamma_i^{(0)^k \frown r(i)} = \gamma_{s'} \gamma_{(0)^k \frown r}$. Hence,
$$\begin{aligned}
 f(s) &= f(s' \frown r) \\
 &= \lim_{n \rightarrow \infty} \gamma_{s' \frown r \upharpoonright n} \cdot x_n \\
 &= \lim_{n \rightarrow \infty} \gamma_{s'} \gamma_{(0)^k \frown r \upharpoonright n} \cdot x_n \quad \text{by the above observation} \\
 &= \gamma_{s'} \lim_{n \rightarrow \infty} \gamma_{(0)^k \frown r \upharpoonright n} \cdot x_n \quad \text{because } \gamma_{s'} \text{ is continuous} \\
 &= \gamma_{s'} f((0)^k \frown r) \\
 &= \gamma_{s'} \gamma_{t'}^{-1} f(t) \quad \text{applying the same equation to } t = t' \frown r.
 \end{aligned}$$
Thus, $\langle f(s), f(t) \rangle \in E_\Gamma^X$.

- Finally, if $\langle s, t \rangle \notin \mathbb{E}_0$ then $\langle f(s), f(t) \rangle \notin R$: assume that $\langle s, t \rangle \notin \mathbb{E}_0$, then for each $N < \omega$ there is some $n > N$ for which $s(n) \neq t(n)$. According with condition 3, the pair $\langle \gamma_{s \upharpoonright n+1} \cdot x_{n+1}, \gamma_{t \upharpoonright n+1} \cdot x_{n+1} \rangle$ belongs to U_n . As $(U_i : i < \omega)$ is decreasing and this happens infinitely-many times, such pair must belong to $\bigcap_{i < \omega} U_i$, so $\langle f(s), f(t) \rangle \in \bigcap_{i < \omega} U_i$. But, $\bigcap_{i < \omega} U_i \cap R = \emptyset$. Hence, $\langle f(s), f(t) \rangle \notin R$. $\square$

Given $R \subseteq X \times X$, define $R_x = \{y \in X : xRy\}$

Theorem 1.19. *Suppose that X is a Polish space, $\langle R_n | n \in \omega \rangle$ is a sequence of F_σ subsets of X^2 , Γ is a group of homeomorphisms of X , and $\mathcal{O} \subseteq X$ is an orbit of Γ with the property that for all $n \in \omega$ and open sets $U \subseteq X$ intersecting $\mathcal{O}$, there are distinct $x, y \in \mathcal{O} \cap U$ with $\mathcal{O} \cap (R_n)_x \cap (R_n)_y = \emptyset$. Then there is a continuous injective homomorphism $f : 2^\omega \rightarrow \bar{\mathcal{O}}$ from $(\mathbb{E}_0, \sim \mathbb{E}_0)$ to $(E_\Gamma^X, \sim \bigcup_{n < \omega} R_n)$.*

Proof. For $x \in \mathcal{O}$ and U a neighborhood of x , there is a different point $y \in U$. So, $\mathcal{O}$ has no isolated points.

We claim that $\bigcup_{n < \omega} R_n$ is meager in $\mathcal{O}$. If U and V are nonempty open subsets of $\mathcal{O}$ (respect to the restricted topology) then $U \cap \bar{\mathcal{O}} \neq \emptyset$ and $V \cap \bar{\mathcal{O}} \neq \emptyset$. So, there are $x, y \in U \cap \mathcal{O}$ different points such that $\mathcal{O} \cap (R_n)_x \cap (R_n)_y = \emptyset$. Thus, for any $z \in V \cap \mathcal{O}$ we have $z \notin \mathcal{O} \cap (R_n)_x \cap (R_n)_y$. This means, either $\langle x, z \rangle \notin R_n$ or $\langle y, z \rangle \notin R_n$. Equivalently, $\sim R_n \cap \mathcal{O} \cap U \times V \neq \emptyset$. Hence, $\sim R_n$ is dense in $\mathcal{O}$, and so R_n is meager in $\mathcal{O}$.

Now apply Theorem 1.18. That is, there exists a continuous injective homomorphism $f : 2^\omega \rightarrow$ from $(\mathbb{E}_0, \sim \mathbb{E}_0)$ to $(E_\Gamma^X, \sim \bigcup_{n < \omega} R_n)$. $\square$

Chapter 2

The Lascar strong type in the Stone space

For this chapter only we will define notions over $\emptyset$ for a complete theory T with monster model $\mathfrak{C}$. It is important to remark that all those notions may be defined over a set of parameters A by understanding them in the complete theory T_A with structure $\mathfrak{C}_A$ obtained from $\mathfrak{C}$ after naming all the elements of A .

Let $Y \subseteq \mathfrak{C}^\alpha$. We say that Y is *pseudo closed* if Y is type definable over some set. We say that Y is *pseudo open* if it is the relative complement (respect to $\mathfrak{C}^\alpha$) of a pseudo closed. *Pseudo F_σ* and *pseudo G_δ* are defined in the obvious way. We say that Y is *closed under the equality of Lascar types* if for all $a \in Y$ such that $a \stackrel{\text{ls}}{=} b$ implies that $b \in Y$. Finally, we say that Y is *d-bounded* if there is $n \in \mathbb{N}$ such that for any $a, b \in Y$, we have $a \stackrel{\text{ls}}{=} b$ if and only if $d(a, b) \leq n$.

Let M be a model of T . Consider the map

$$\zeta_{\alpha, M} : \mathfrak{C}^\alpha \longrightarrow S_\alpha(M)$$

which sends every α -tuple a to the type $tp(a/M)$. We will omit α and M when it is not ambiguous from the context.

We have mention that in a Polish space X , a subspace is Polish if and only if it is a G_δ subset of X . The question about the Borel cardinality of $\stackrel{\text{ls}}{=}$ on $\mathfrak{C}^\alpha$ may be generalized by restricting equality of Lascar types over a pseudo G_δ set. Thus, we are interested in knowing what conditions we need to state that the mapping $\zeta_{\alpha, M}$ sends pseudo G_δ to G_δ subsets of $S(M)$.

Let $Y \subseteq \mathfrak{C}^\alpha$ be pseudo closed, this is, Y is type definable by a partial type Γ over a set A . The direct image of Y by $\zeta_{\alpha, M}$ is denoted by Y_M . Particularly, observe that $Y_{\mathfrak{C}} = \{p \in S(\mathfrak{C}) : \Gamma \subseteq p\} = [\Gamma]$. Thus, if $[\Gamma] \subseteq \bigcup_{i \in I} [\varphi_i]$, then $\Gamma \cup \{\neg \varphi_i : i \in I\}$ is

inconsistent. Using compactness, there is a natural number n such that $[\Gamma] \subseteq \bigcup_{i < n} [\varphi_i]$. In other words, every covering of $Y_{\mathfrak{C}}$ has a finite subcovering.

It is not hard to check that the restriction mapping γ from $S(\mathfrak{C})$ to $S(M)$ which sends every complete type p to $p \upharpoonright M$ is continuous, and that the direct image of $Y_{\mathfrak{C}}$ by γ is Y_M . Assume that Y_M is covered by open subsets O_i of $S(M)$, for $i \in I$. Then, $Y_{\mathfrak{C}}$ is covered by the open subsets $\gamma^{-1}[O_i]$, for $i \in I$, and in consequence there is a finite subcovering of $Y_{\mathfrak{C}}$. Hence, Y_M is compact. Since $S(M)$ is Hausdorff, Y_M is closed in $S(M)$.

Therefore, $\zeta_{\alpha, M}$ sends pseudo closed of $\mathfrak{C}^\alpha$ to closed subsets of $S(M)$. Furthermore, $\zeta_{\alpha, M}$ sends pseudo F_σ of $\mathfrak{C}^\alpha$ to F_σ subsets of $S(M)$. However, this is not true for pseudo G_δ , unless such sets are closed under the Lascar equivalence.

Remark 2.1. Suppose that $Y \subseteq \mathfrak{C}^\alpha$ is pseudo G_δ closed under equality of Lascar types. Let $p \in S(M)$ be such that $p = tp(a/M)$, for some $a \in \mathfrak{C}^\alpha \setminus Y$. If there is a' in Y for which $tp(a'/M) = p$, then $a \stackrel{\text{ls}}{=} a'$. This leads a contradiction. Thus, $\gamma_{\alpha, M}[\mathfrak{C}^\alpha \setminus Y] = S_\alpha(M) \setminus Y_M$ is an F_σ set. Hence, Y_M is a G_δ subset of $S(M)$.

Definition 2.2. Let M be a model of T . For $p, q \in S(M)$, we write $p \stackrel{\text{ls}}{=} q$ if there are tuples a and b for which $a \models p$, $b \models q$ and $a \stackrel{\text{ls}}{=} b$. Similary, we write $p \stackrel{\text{kp}}{=} q$ if there are tuples a and b for which $a \models p$, $b \models q$ and $a \stackrel{\text{kp}}{=} b$. Furthermore, we define $d(p, q) = \min\{n : \exists a \models p \exists b \models q \ d(a, b) \leq n\}$.

Notice that if $Y \subseteq \mathfrak{C}^\alpha$ is closed under equality of Lascar types, then for any model M we have that Y_M is closed under the equivalence relation $\stackrel{\text{ls}}{=}$ on $S(M)$. Assume that $p \stackrel{\text{ls}}{=} q$ and $q \in Y_M$, there are tuples a, b such that $a \models p$ and $b \models q$ such that $a \stackrel{\text{ls}}{=} b$. Since $b \in Y$ implies that $a \in Y$, we obtain that $p \in Y_M$.

Lemma 2.3 (Remark 2.2, [16]). *Let M be a model. For any $p, q \in S(M)$, the following are equivalent:*

1. $p \stackrel{\text{ls}}{=} q$
2. For all $a \models p$ and $b \models q$, $a \stackrel{\text{ls}}{=} b$.

Furthermore, $\stackrel{\text{ls}}{=}$ is an equivalence relation on the Stone space $S(M)$.

Proof. From left to right. Let $a \models p$ and $b \models q$ be such that $a \stackrel{\text{ls}}{=} b$ and consider any $a' \models p$ and $b' \models q$. Then $a \equiv_M a'$ and $b \equiv_M b'$, this means that $a \stackrel{\text{ls}}{=} a'$ and $b \stackrel{\text{ls}}{=} b'$. Hence, $a' \stackrel{\text{ls}}{=} b'$. The converse is clear. $\square$

Let M be a model. The mapping that assigns each pair $(tp(a/M), tp(b/M))$ to $tp(ab/M)$ is a well-defined and continuous mapping from $S(M) \times S(M)$ to $S(M)$. For each natural number n , the collection $\{(a, b) \in \mathfrak{C}^\alpha \times \mathfrak{C}^\alpha : d(a, b) \leq n\}$ is clearly type definable (over

$\emptyset$) by $nc^n(x, y)$ and, consequently, the set $\{tp(ab/M) \in S(M) : d(a, b) \leq n\}$ is closed in $S(M)$. Hence, the set $F_n := \{(p, q) \in S(M) \times S(M) : d(p, q) \leq n\}$ is closed in the product topology $S(M) \times S(M)$, for all $n \in \mathbb{N}$.

Since equality of Lascar strong types is determined by an infinite disjunction of type definables (over $\emptyset$), we obtain that for each pair $p, q \in S(M)$,

$$p \stackrel{\text{ls}}{=} q \text{ if and only if there exists a natural number } n \text{ for which } d(p, q) \leq n.$$

In other words, $\stackrel{\text{ls}}{=}$ is an F_σ equivalence relation on $S(M)$, and therefore it is a Borel equivalence relation on the Stone space $S(M)$. Using a similar argument, one can conclude that $\stackrel{\text{kp}}{=}$ is a closed equivalence relation on $S(M)$. In particular, $\stackrel{\text{kp}}{=}$ -equivalence classes are closed, and so compact subsets of $S(M)$, (see Remark 2.5, [16]).

In order to understand the Borel cardinality of the Lascar equivalence relation defined in two Polish spaces $S(M)$ and $S(N)$ it is important to recall the following fact:

(Fact 1.7, [16]). Let E and F be Borel equivalence relations on two Polish spaces X and Y , respectively, and $f : X \rightarrow Y$ be a continuous surjective function such that $\langle x_1, x_2 \rangle \in E$ iff $\langle f(x_1), f(x_2) \rangle \in F$, for all $x_1, x_2 \in X$. Then $E \sim_B F$.

Let α be a countable ordinal. For M and N countable models, we may assume that N is an elementary substructure of M . The mapping that assigns each complete type p to $p \upharpoonright N$ is a continuous surjection from $S(M)$ to $S(N)$. In fact, if $a \models p$ then $a \models p \upharpoonright N$. According with the fact mentioned above, we obtain that the Lascar equivalence relation $\stackrel{\text{ls}}{=}$ on $S(M)$ has the same Borel cardinality that the Lascar equivalence relation $\stackrel{\text{ls}}{=}$ on $S(N)$.

For a complete type of p over M , let's denote as $[p]_{kp}$ the $\stackrel{\text{kp}}{=}$ -equivalence class of p in $S(M)$. Thus, using a similar argument as before one can conclude the following corollary:

Corollary 2.4. *Let M and N be countable models. For a tuple a we have*

$$\stackrel{\text{ls}}{=} \upharpoonright [tp(a/M)]_{kp} \sim_B \stackrel{\text{ls}}{=} \upharpoonright [tp(a/N)]_{kp}$$

Corollary 2.5. *Let $Y \subseteq \mathfrak{C}^\alpha$ be a pseudo G_δ closed under equality of Lascar types. Then,*

$$\stackrel{\text{ls}}{=} \upharpoonright Y_M \sim_B \stackrel{\text{ls}}{=} \upharpoonright Y_N$$

Proof. Using Remark 2.1, if Y is a pseudo G_δ closed under Lascar types, Y_M and Y_N are G_δ sets in $S(M)$ and $S(N)$, respectively. Hence, Y_M and Y_N are Polish spaces accordingly with Theorem 1.5. Finally, argue as before. $\square$

Theorem 2.6. *Let T , M and α be countable. Let $Y \subseteq \mathfrak{C}^\alpha$ be a pseudo G_δ closed under equality of Lascar types.*

If there is some $p \in Y_M$ such that for every neighborhood $U \subseteq S(M)$ of p and natural number k there exists a strong automorphism σ such that:

1. σ fixes M setwise.
2. $p^\sigma \in U$ and $d(p^\sigma, p) > k$.

Then there is a continuous and injective homomorphism $f : 2^\omega \longrightarrow Y_M$ from $\mathbb{E}_0$ to $\equiv^{\text{ls}} \upharpoonright Y_M$. In particular, $\equiv^{\text{ls}} \upharpoonright Y_M$ is nonsmooth.

Proof. As Y is pseudo a G_δ , Y_M is a G_δ set in $S(M)$, and so it is Polish by Theorem 1.5; we also have, by Condition 1, that $p^\sigma \in Y_M$, for all $p \in Y_M$. In order to apply Theorem 1.19, we want to find a sequence $\langle R_n \mid n \in \omega \rangle$ of F_σ subsets of Y_M , a group of homeomorphisms Γ of Y_M , and an orbit $\mathcal{O} \subseteq Y_M$ with the property that for all $n \in \omega$ and open sets $U \subseteq Y_M$ intersecting $\mathcal{O}$, there are distinct $x, y \in \mathcal{O} \cap U$ such that $\mathcal{O} \cap (R_n)_x \cap (R_n)_y$ is empty.

Take the sequence of closed sets of Y_M defined by $R_n = F_n \cap Y_M \times Y_M$, where F_n is the set of pairs $(p, q) \in S(M) \times S(M)$ such that $d(p, q) \leq n$, and Γ is the group of automorphisms of Y_M induced by $\text{Aut}f(\mathfrak{C})$. We claim that the orbit $\mathcal{O}$ of the type p as in the hypothesis of this theorem has the property as in Theorem 1.19.

Let $B_\varepsilon(p)$ be the ball centered in p with radius $\varepsilon < 1$ in Y_M . For V an open subset of Y_M clearly we have that $V \cup B_\varepsilon(p)$ is a neighborhood of p . Then for $k = 1$ there is $\sigma \in \text{Aut}f(\mathfrak{C})$ such that $p^\sigma \in V \cup B_\varepsilon(p)$ and $d(p, p^\sigma) > 1$. Hence, $p^\sigma \in V$. That means $\mathcal{O}$ is dense in Y_M . If we consider $U = \sigma^{-1}[V]$ and a strong automorphism τ for which $p^\tau \in U$ and $d(p, p^\tau) > 2n$, then $d(p^\sigma, p^{\sigma\tau}) = d(p, p^\tau) > 2n$. Hence, $p^\sigma \neq p^{\sigma\tau}$ and $\mathcal{O} \cap (R_n)_{p^\sigma} \cap (R_n)_{p^{\sigma\tau}}$ is clearly empty.

Notice that $\equiv^{\text{ls}} \upharpoonright Y_M$ is $\bigcup_{n < \omega} R_n$, because $\equiv^{\text{ls}} \upharpoonright S(M)$ is $\bigcup_{n < \omega} F_n$. Applying Theorem 1.19, there is a continuous injective homomorphism $f : 2^\omega \longrightarrow Y_M$ from $\mathbb{E}_0$ to $\equiv^{\text{ls}} \upharpoonright Y_M$. In other words, $\mathbb{E}_0 \sqsubseteq_c^{\text{ls}} \equiv^{\text{ls}} \upharpoonright Y_M$, and therefore $\equiv^{\text{ls}} \upharpoonright Y_M$ is nonsmooth. $\square$

Chapter 3

Small Case

Kaplan, Miller and Simon proved what they called in [11] *Main Theorem A* stated as follows:

Main theorem A. *Suppose T and α are countable, and let Y be a pseudo G_δ of $\mathfrak{C}^\alpha$ closed under $\stackrel{\text{ls}}{=}$. If for some $a \in Y$, its Lascar type $[a]_{\text{ls}}$ is not d -bounded, then $\stackrel{\text{ls}}{=} \upharpoonright Y$ is nonsmooth.*

At the end of Chapter 2 we proved Theorem 2.6. This theorem is certainly a good approach in order to analyze the Borel cardinality of the Lascar relation on Polish spaces. As a matter of fact, Theorem 2.6 opens the possibility to prove *Main Theorem A* for the cases in which T is small or countable. We will mainly be concerned with these two cases during this chapter and the next one. Nevertheless, I just will mention a simplified version of the case in which T has any size. Regretfully, I could not extend more on this result which is interesting because it carries with itself four important results that have been proved in the literature with, sometimes, powerful tools.

Main theorem B. *[Simplified version] Suppose T is a complete first-order theory, α is a small ordinal. Suppose Y is a pseudo G_δ of $\mathfrak{C}^\alpha$ closed under $\stackrel{\text{ls}}{=}$ and for some $a \in Y$ its Lascar type $[a]_{\text{ls}}$ is not d -bounded. Then the quotient $Y / \stackrel{\text{ls}}{=}$ has at least $2^{\aleph_0}$ elements.*

Definition 3.1. We say that a countable model M is *nice* if:

1. for any pair a and b of finite tuples in M such that $a \stackrel{\text{ls}}{=} b$, there is an automorphism σ of M that can be extended to some $\tau \in \text{Aut}f(\mathfrak{C})$ mapping a to b .
2. for each finite tuple a in M and natural number n , if there exist c_1, c_2 such that $c_1 \stackrel{\text{ls}}{=} a \stackrel{\text{ls}}{=} c_2$ and $d(c_1, c_2) > n$, then there are such tuples c_1, c_2 inside M .

3. for all a and b finite tuples of the same length in M , another finite tuple a' in M (possibly with different length) and a natural number n , if there exists some b' such that $d(aa', bb') \leq n$, then there is one such tuple b' inside M .

Lemma 3.2. *Let A be a countable set and $\{\sigma_i : i < \omega\}$ be a sequence of strong automorphisms. There exists a countable model M containing A such that σ_i fixes M setwise, for all $i < \omega$*

Proof. Let M_0 be a countable model containing A .

Recursively, assume that M_n is a countable model. Take M_{n+1} to be a countable model extending M_n and containing the countable set $\bigcup_{i \leq n} \sigma_i[M_n] \cup \bigcup_{i \leq n} \sigma_i^{-1}[M_n]$.

For $M = \bigcup_{i < \omega} M_i$, it is clear that M is a model. Fix $i < \omega$ and $a \in M$. Since $a \in M_j$, for some $j < \omega$, then there is $k \geq i, j$ such that M_{k+1} contains $\sigma_i[M_k] \cup \sigma_i^{-1}[M_k]$. Hence, σ_i fixes M setwise. $\square$

Lemma 3.3. *T has nice models. Moreover, for every countable set A , there is a nice model containing A .*

Proof. Let M_0 be a countable model which contains A .

Proceeding by induction, assume that M_k is given. First, for each finite tuple $a \in M_k$ and natural number n , pick c_1 and c_2 in $[a]_{l_s}$ such that $d(c_1, c_2) > n$, whenever they exist. It is clear that the set B containing such tuples c_1 and c_2 is countable. Similarly, for finite tuples $a, b \in M_k$, another tuple $a' \in M_k$ and natural number n , pick a tuple b' (with the same length as a') such that $d(aa', bb') \leq n$, whenever it exists. It is also clear that the set C containing such tuples b' is countable. M_{k+1} will be the model containing $M_k \cup B \cup C$ obtained using Lemma 3.2. $\square$

Theorem 3.4. *Let α be a countable ordinal, and $Y \subseteq \mathfrak{C}^\alpha$ be a pseudo G_δ closed under $\stackrel{\text{ls}}{=}$. If there is $a \in Y$ such that $[a]_{l_s}$ is not d -bounded and the Lascar strong type of every finite subtuple of a is d -bounded. Then $\stackrel{\text{ls}}{=} \restriction Y$ is nonsmooth.*

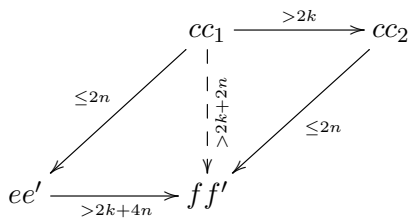
Proof. Let M be a nice model containing a and let $p = tp(a/M)$. By Remark 2.1, Y_M is G_δ , and hence Y_M is a Polish space. Besides, if σ is a strong automorphism that fixes M setwise, then $p^\sigma \in S(M)$. Furthermore, $p \stackrel{\text{ls}}{=} p^\sigma$, because $a \stackrel{\text{ls}}{=} \sigma(a)$. Since Y is closed under $\stackrel{\text{ls}}{=}$, by Remark 2.1, so Y_M is closed under $\stackrel{\text{ls}}{=}$. Hence, σ fixes Y_M setwise.

Let U be an open subset of $S(M)$ containing p . We may assume that $U = [\varphi]$, for some $\varphi = \varphi(x) \in \mathcal{L}(M)$. So, $\varphi \in p$ and $M \models \varphi(a)$. Let $c = (a_{i_1}, \dots, a_{i_m})$ be a finite subtuple

of a for which $M \models \varphi(c)$. Denote with $x = c$ the formula $x_{i_1} = a_{i_1} \wedge \dots \wedge x_{i_m} = a_{i_m}$. Since, $x = c \vdash \varphi(x)$ and $p \in [x = c]$, we may assume $U = [x = c]$.

Fix a natural number k . By hypothesis, the Lascar strong type of c is d -bounded by some natural number n . Nevertheless, $[a]_{ls}$ is not d -bounded. So, there is some finite subtuple c' of a different from c such that $[cc']_{ls}$ has a diameter greater than $2k + 4n$. This means that there are tuples $ee', ff' \in [cc']_{ls}$ for which $d(ee', ff') > 2k + 4n$. Since M is nice, we can chose such tuples ee', ff' in M .

Using niceness of M , there are c_1, c_2 tuples of M such that $d(ee', cc_1) \leq 2n$ and $d(ff', cc_2) \leq 2n$.



Applying twice Triangular Inequality, we obtain that $d(cc_1, cc_2) > 2k$.

But $2k < d(cc_1, cc_2) \leq d(cc', cc_1) + d(cc', cc_2)$, and then $d(cc', cc_1) > k$ or $d(cc', cc_2) > k$.

Say $d(cc', cc_1) > k$

As c_1 was chosen, $cc' \stackrel{ls}{\equiv} cc_1$. Let σ be in $\text{Aut}f(\mathfrak{C})$ mapping cc' to cc_1 . Notice that σ fixes c , so $x = c \in p^\sigma$. Furthermore, since $a \models p$ and $\sigma(a) \models p^\sigma$ where $\sigma(a)$ contains the subtuple cc_1 , then $d(p, p^\sigma) > k$. Apply now Theorem 2.6. $\square$

Definition 3.5. A theory T is *small* if $|S_n(\emptyset)|$ is countable for all $n < \omega$.

It is not hard to verify that requiring $S_n(\emptyset)$ to be countable is equivalent to requiring $S_n(A)$ to be countable, for A finite or $n = 1$. Something more interesting is that T is small whenever it has a saturated countable model; there is a proof of that in [3]. Additionally, if T is a small theory then there is a countable language $\mathcal{L}' \subseteq \mathcal{L}$ such that for every $\varphi(x) \in \mathcal{L}$ there is some $\varphi'(x) \in \mathcal{L}'$ equivalent with $\varphi(x)$, that is, $\varphi(x) \equiv \varphi'(x)$ (see Remark 4.25, [3]). Hence, every small theory can be taken as a countable theory. Nonetheless, Newelski proved in [19] that equality of Lascar strong types is type-definable if and only if for some $n < \omega$ it is defined by $nc^n(x, y)$ (see also [3]). Moreover, in the case that T is small, for any finite tuple a , its Lascar type is the same as its Kim-Pillay type. Applying Theorem 3.4, we have proved the following corollary.

Corollary 3.6. Let T be a small theory and α be a countable ordinal. If $Y \subseteq \mathfrak{C}^\alpha$ as in Theorem 3.4, then $\stackrel{ls}{\equiv} \upharpoonright Y$ is nonsmooth.

Chapter 4

The countable case

Kaplan, Miller and Simon introduced the notion of *properness* in order to find a type p whose orbit under the group of strong automorphisms is sufficiently rich (as in Theorem 2.6), in the case we have an unbounded Lascar type. During this chapter we will take care of the theory developed in [11] for this matter. We will see in detail how Kaplan, Miller and Simon constructed formula-by-formula such p making sure of having room to go on at each stage, in a way so that p has conjugates with distance from each other as large as we want.

For this chapter only we will assume that T and α are countable. Many of the notions that we will see during this chapter, depend on translations of formulas -or partial types- in $\mathcal{L}(\mathfrak{C})$. In order to avoid extensive notation, for this chapter only we will assume that all the formulas -and partial types- are over $\mathfrak{C}$. So, for a formula $\varphi = \varphi(x)$ by φ^σ we will mean $\varphi(x, \sigma(a))$, for some tuple a in $\mathfrak{C}$ and automorphism σ . Similarly for types.

Let $C \subseteq \mathfrak{C}^\alpha$ be a fixed Lascar strong type. We say that a formula $\varphi = \varphi(x)$ is called *C-generic*, or simply *generic*, if there are strong automorphisms $\sigma_0, \dots, \sigma_{n-1}$ such that $C \subseteq \bigcup_{i < n} \varphi(\mathfrak{C})^{\sigma_i}$. A formula φ is said to be *C-weakly generic*, or simply *weakly generic*, if there is a *C*-nongeneric formula $\psi = \psi(x)$ such that $\varphi \vee \psi$ is *C-generic*. A partial type $p = p(x)$ is said to be *C-generic*, or simply *generic*, if all the formulas that p implies are generic. Finally, p is called *C-weakly generic*, or simply *weakly generic*, if all the formulas that p implies are weakly generic.

Observation 4.1. Suppose that φ_1 and φ_2 are two non-weakly generic formulas. On one the hand, assume that $\varphi_1 \vee \varphi_2$ is weakly generic. Let ψ be a nongeneric formula such that $\varphi_1 \vee \varphi_2 \vee \psi$ is generic. Since, φ_2 is non-weakly generic, then $\varphi_2 \vee \psi$ is non generic. This implies that φ_1 is weakly generic, which is a contradiction. So, $\varphi_1 \vee \varphi_2$ is non-weakly generic. On the other hand, let $\varphi_1 \vdash \varphi_2$. It is clear that if φ_1 is weakly generic, φ_2 is

weakly generic. Therefore, sets defined by formulas which are non-weakly generic form an ideal.

Definition 4.1. Suppose p is weakly generic. We say that p is *C-proper*, or simply *proper*, if there is a nongeneric formula ψ such that $\varphi \vee \psi$ is generic, for every formula φ implied by p .

Observation 4.2. A formula that implies a contradiction, say $\forall x(\neg x = x)$, is nongeneric. Hence, any generic partial type is proper.

Proposition 4.2. *If p_1 and p_2 are partial types over the same set such that $p_1 \vdash p_2$ and p_1 is proper, then p_2 is proper.*

Proof. Since every formula φ implied by p_2 is also implied by p_1 , the formula that witnesses the properness of p_1 works for p_2 . $\square$

Even though, *properness* only concerns partial types, Proposition 4.2 assures us that the definition can be transferred in a natural way to pseudo closed. Namely, if F is a pseudo closed defined by some partial types p_1 and p_2 over the same set, then p_1 is proper if and only if p_2 is proper.

Observation 4.3. Let p_1 and p_2 be nonproper partial types. We may assume that p_1 and p_2 are over the same set A , where A is the union of the parameters of p_1 and p_2 . Let p be the type $\{\varphi_1 \vee \varphi_2 : \varphi_1 \in p_1, \varphi_2 \in p_2\}$. If p were proper, let ψ be a formula witnessing it. In particular, for ψ there are $\varphi_1 \in p_1$ and $\varphi_2 \in p_2$ such that $\varphi_1 \vee \psi$ and $\varphi_2 \vee \psi$ are not generic. But, as we chose ψ , $\varphi_1 \vee \varphi_2 \vee \psi$ is generic, which is a contradiction. Therefore, in conjunction with Proposition 4.2, nonproper pseudo closed form an ideal.

Lemma 4.3. *Let p be a proper partial type. Then for every formula $\varphi = \varphi(x)$, either $p \cup \{\varphi\}$ or $p \cup \{\neg\varphi\}$ is proper.*

Proof. Let ψ be a formula witnessing the properness of p . Assume that $p \cup \{\varphi\}$ and $p \cup \{\neg\varphi\}$ are nonproper, then $q = p \cup \{\varphi \vee \neg\varphi\}$ is nonproper, which is a contradiction since $\varphi \vee \neg\varphi \vee \psi$ is generic. $\square$

Lemma 4.4. *Let p be a partial proper type witnessed by a formula ψ . For $\varphi_0, \dots, \varphi_{n-1}, \chi$ formulas such that $\bigvee_{i < n} \varphi_i \vee \chi$ covers C and $\psi \vee \chi$ is nongeneric, there is $i < n$ for which $p \cup \{\varphi_i\}$ is proper.*

Proof. Assume that $p \cup \{\varphi_i\}$ is nonproper, for each $i < n$. Because p is proper, each φ_i is non-weakly generic, and so $\bigvee_{i < n} \varphi_i$ is non-weakly generic. Since $\psi \vee \chi$ is nongeneric, χ is nongeneric. This is a contradiction with the assumption $\bigvee_{i < n} \varphi_i \vee \chi$ covers C . $\square$

Proposition 4.5. *For a partial proper type $p = p(x)$ and a formula $\varphi \in p$, there are strong automorphisms $\tau_0, \dots, \tau_{n-1}$ such that for any $\sigma \in \text{Autf}(\mathfrak{C})$ there is some $i < n$ for which $p \cup \{\varphi^{\sigma\tau_i}\}$ is proper.*

Proof. Let ψ be a formula witnessing the properness of p and let $\tau_0, \dots, \tau_{n-1} \in \text{Autf}(\mathfrak{C})$ be such that $\bigvee_{i < n} (\varphi \vee \psi)^{\tau_i}$ covers C . Let σ be a strong automorphism. Since σ fixes C setwise, $\bigvee_{i < n} (\varphi \vee \psi)^{\sigma\tau_i} = \bigvee_{i < n} \varphi^{\sigma\tau_i} \vee \bigvee_{i < n} \psi_i^{\sigma\tau_i}$ covers C . Denote by χ the formula $\bigvee_{i < n} \psi^{\sigma\tau_i}$. Observe that it is nongeneric because ψ and hence $\psi^{\sigma\tau_i}$ is nongeneric. Applying Lemma 4.4, we obtain what is desired. $\square$

Proposition 4.6. *For any $a \in C$, the partial type $nc(x, a)$ is proper.*

Proof. To show a contradiction, assume that $nc(x, a)$ is nongeneric. Let $\varphi(x, a)$ be a nongeneric formula in $nc(x, a)$, that means that for every $a_0 \stackrel{\text{ls}}{\equiv} \dots \stackrel{\text{ls}}{\equiv} a_{n-1} \stackrel{\text{ls}}{\equiv} a$, there exists a' such that $a' \vdash \bigwedge_{i < n} \neg\varphi(x, a_i)$. But this is a contradiction, because φ is finite. Hence, $nc(x, a)$ is generic, and so is proper. $\square$

Theorem 4.7. *If $Y \subseteq \mathfrak{C}^\alpha$ is pseudo G_δ closed under $\stackrel{\text{ls}}{\equiv}$ such that for some $a \in Y$, $[a]_{\text{ls}}$ is not d -bounded, then $\stackrel{\text{ls}}{\equiv} \upharpoonright Y$ is nonsmooth.*

Proof. The strategy of the proof consists in finding a countable model M of T and a set $Y_M \subseteq S(M)$ in accord with Theorem 2.6. Without loss of generality, we may take clopen sets $[\varphi]$ of $S(M)$ instead of any arbitrary open set U of $S(M)$. Thus, we can reformulate the conditions of Theorem 2.6 as follows:

There is some $p \in Y_M$ such that for every formula $\varphi \in p$ and natural number N , there exists a strong automorphism σ fixing M setwise such that $\varphi \in p^\sigma$ and $d(p, p^\sigma) > N$.

For purposes of this proof, when we write *proper* we will mean $[a]_{\text{ls}}$ -proper. Also, $q = q(x, a)$ denotes the type $nc(x, a)$.

Claim 1. There is a countable sequence of strong automorphisms $(\sigma_i : i < \omega)$, a countable $\subseteq$ -increasing chain $(A_i : i < \omega)$ of finite subsets of $\mathfrak{C}$, and a countable $\subseteq$ -increasing chain of partial types $(p_i : i < \omega)$ over $\mathfrak{C}$ with the following properties:

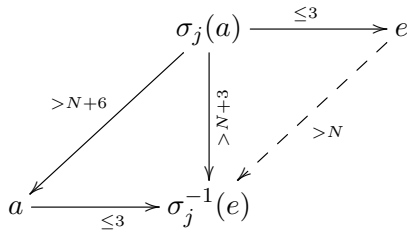
1. For all $i < \omega$ and consistent formula $\varphi \in \mathcal{L}_1(A_i)$, there is $j > i$ such that for some $c \in A_j$, $\models \varphi(c)$.
2. For all $i, k < \omega$ there exists some $j > i$ such that A_j contains $\sigma_k[A_i]$ and $\sigma_k^{-1}[A_i]$.
3. Every p_i is a finite partial type over A_i such that $p_i \cup q$ is proper.

4. For all $i < \omega$ and formula $\varphi \in \mathcal{L}_\alpha(A_i)$, there is some $j > i$ such that p_j contains either φ or $\neg\varphi$.
5. For every $i < \omega$, formula $\varphi \in p_i$ and natural number N , there are $k > j > i$ such that $d(\sigma_j(a), a) > N$ and $\varphi^{\sigma_j^{-1}} \in p_k$.

Assume that Claim 1 is true. First, take $M = \bigcup_{i < \omega} A_i$, and observe that it is a countable model of T , using 1. Hence, $S(M)$ is a Polish space, and so Y_M is a Polish space. Furthermore, for any σ_i in the sequence and $a \in M$, there exists $j > i$ such that $\sigma_i(a) \in A_j$, then σ_i fixes M setwise.

Secondly, for $p = \bigcup_{i < \omega} p_i$ is clear that $p \in S(M)$, using 4. Moreover, notice that for all $c \models p$ and $c' \models p \cup q$, we have $d(c, c') \leq 2$ and $d(c', a) \leq 1$; and because of Triangle Inequality, $d(a, c) \leq 3$. As Y is closed under $\stackrel{\text{ls}}{=}$, we obtain that p belongs to Y_M .

Let $\varphi \in p$ and N be a natural number. Let σ_j be such that $d(\sigma_j(a), a) > N + 6$ and $\varphi^{\sigma_j^{-1}} \in p$, as in 5. We noticed that σ_j fixes M setwise. Moreover, $\varphi \in p^{\sigma_j}$, since $\varphi^{\sigma_j^{-1}} \in p$.



Let $e \in \mathfrak{C}$ be such that $e \models p^{\sigma_j}$. Because $\sigma_j^{-1}(e) \models p$, $d(a, \sigma_j^{-1}(e)) \leq 3$. Since σ_j preserves distances, $d(\sigma_j(a), e) \leq 3$. Using Triangular Inequality twice, we obtain that $d(\sigma_j^{-1}(e), e) > N$. Therefore, $d(p, p^\sigma) > N$

□

Proof of Claim 1. Let $A_0 = \emptyset$ and $p_0 = \emptyset$.

Assume that A_k is given and let e_k be an injective mapping from $\mathcal{L}(A_k)$ to a coinfinite subset of $\mathbb{N}$. Let $j \leq k$ be in range of e_k . If $\varphi = \varphi(x, b) \in \mathcal{L}(A_k)$ is a consistent formula which is the preimage of j under e_k and x is a single variable, pick an element of the monster model c_j witnessing $\varphi(x, b)$. Define

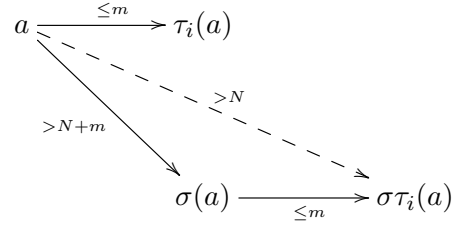
$$A_{k+1} := A_k \cup \{c_j : j \leq k\} \cup \bigcup_{i < k} \sigma_i[A_k] \cup \bigcup_{i < k} \sigma_i^{-1}[A_k]$$

A_k is finite. Define e_{k+1} as an injective mapping from $\mathcal{L}(A_{k+1})$ to a coinfinite subset of $\mathbb{N}$ compatible with e_k , i.e., such that $e_k \subseteq e_{k+1}$.

Now suppose that p_k is given. We want to construct p_{k+1} and a finite number of $\sigma_k^l \in \text{Aut}f(\mathfrak{C})$. Let p' be the partial type obtained by applying at most k -times Lemma 4.3 to formulas $\varphi \in \mathcal{L}(A_k)$ that are preimages of some $j < k$ under e_k such that $p' \cup q$ is proper.

Let f_k be an injective function from $p_k \times \mathbb{N}$ to a coinfinite subset of $\mathbb{N}$ and let $\langle \varphi, N \rangle$ be the preimage of some $j < k$ under f_k (whenever it exists).

Since $p' \cup q$ is proper and $\varphi \in p' \cup q$, there are strong automorphisms $\tau_0, \dots, \tau_{n-1}$ as in Proposition 4.5. Also, by unboundedness of $[a]_{ls}$, there is a $\sigma \in \text{Aut}f(\mathfrak{C})$ such that $d(a, \sigma(a)) > N + m$, where $m \geq \max\{d(a, \tau_i(a)) : i < n\}$. Therefore, $d(\sigma(a), \sigma\tau_i(a)) \leq m$.



Hence, $d(a, \sigma\tau_i(a)) > N$, for each $i < n$. Proposition 4.5 assures us that there is $i < n$ such that $p' \cup q \cup \{\varphi^{\sigma\tau_i}\}$ is proper. Define $\sigma_k^j = \sigma\tau_i$, for such i given by Proposition 4.5. Repeating this process at most k -times, starting from $p' \cup q$, we can obtain a finite partial type p_{k+1} and at most k -many strong automorphisms σ_k^j with the required conditions. $\square$

One can zoom out and ask what would happen if instead of restricting $\overset{ls}{\equiv}$ to a Lascar type we restrict it to a Kim-Pillay type. After proving this theorem, the answer follows straightforward just by noticing that if K is a d -unbounded Kim-Pillay type, then it must contain a d -unbounded Lascar type; otherwise all its classes will have the same bound, since they are conjugates. Using Theorem 4.7, the restriction of Lascar equivalence relation $\overset{ls}{\equiv} \upharpoonright K$ is nonsmooth.

Likewise, if $\overset{ls}{\equiv}$ is not pseudo closed then $\overset{ls}{\equiv}$ does not coincide with $\overset{kp}{\equiv}$, so there is a Kim-Pillay type K such that $\overset{ls}{\equiv} \upharpoonright K$ is not trivial, and hence $\overset{ls}{\equiv} \upharpoonright K$ is nonsmooth and neither $\overset{ls}{\equiv}$. Using Proposition 1.14 we obtain that whenever T and α are countable, $\overset{ls}{\equiv}$ is closed if and only if it is smooth.

Bibliography

- [1] JECH, THOMAS. *Set theory*. The third millennium edition, revised and expanded. Springer Monographs in Mathematics. Springer-Verlag, Berlin, 2003.
- [2] BECKER, HOWARD; KECHRIS, ALEXANDER S. *The descriptive set theory of Polish group actions*. London Mathematical Society *Lecture Note Series*, 232. Cambridge University Press, Cambridge, 1996.
- [3] CASANOVAS, ENRIQUE. *Simple theories and hyperimaginaries*. *Lecture Notes in Logic*, 39. Association for Symbolic Logic, Chicago, IL; Cambridge University Press, 2011.
- [4] CASANOVAS, ENRIQUE; LASCAR, DANIEL; PILLAY, ANAND; ZIEGLER, MARTIN. *Galois groups of first order theories*. *J. Math. Log.* 1 (2001), no. 2, 305–319.
- [5] COSKEY, SAMUEL; SCHNEIDER, SCOTT. *Borel cardinal invariant properties of countable Borel equivalence relations*, [ArXiv](#), 2011.
- [6] R. DOUGHERTY; S. JACKSON; A.S. KECHRIS. *The structure of hypernite Borel equivalence*. Second revised edition. *Trans. Amer. Math. Soc.*, 341 (1), 193–225, 1994.
- [7] FRAENKEL, ABRAHAM; BAR-HILLEL, YEHOASHUA; LEVY, AZRIEL. *Foundations of set theory*. Second revised edition. *Studies in Logic and the Foundations of Mathematics*, Vol. 67. North-Holland Publishing Co., Amsterdam-London, 1973.
- [8] HARRINGTON, LEO A.; KECHRIS, ALEXANDER. S.; LOUVEAU, ALAIN. *A Glimm-Effros dichotomy for Borel equivalence relations*. *J. Amer. Math. Soc.* 3 (1990), no. 4, 903–928.
- [9] IVANOV, ALEXANDRE A. *A countably categorical theory which is not G-compact*. *Siberian Adv. Math.* 20 (2010), no. 2, 75–82.
- [10] KAPLAN, ITAY; MILLER, BENJAMIN. *An embedding theorem of $\mathbb{E}_0$ with model theoretic applications*, [ArXiv](#). (2013).

- [11] KAPLAN, ITAY; MILLER, BENJAMIN; SIMON, PIERRE. *The Borel cardinality of Lascar Strong types*, [ArXiv](#).
- [12] KECHRIS, ALEXANDER S. *Classical descriptive set theory*. Graduate Texts in Mathematics, 156. Springer-Verlag, New York, 1995. ISBN: 0-387-94374-9
- [13] KEREN, AVIV. *Equivalence relations and topological automorphism groups in Simple theories*. Master thesis. [Einstein Institute of Mathematics](#), Hebrew University. 2007.
- [14] KIM, BYUNGHAN. *Recent results on simple first order theories. Model theory of groups and automorphism groups*. (Blaubeuren, 1995), 202–212, London Math. Soc. Lecture Note Ser., 244, Cambridge Univ. Press, Cambridge, 1997.
- [15] KIM, BYUNGHAN. *A note on Lascar strong types in simple theories*. J. Symbolic Logic 63 (1998), no. 3, 926–936.
- [16] KRUPIŃSKI, KRZYSZTOF; PILLAY, ANAND; SOLECKI, SŁAWOMIR. *Borel equivalence relations and Lascar strong types*. J. Math. Log. 13 (2013), no. 2, 37 pp.
- [17] MOSCHOVAKIS, YIANNIS N. *Descriptive set theory*. Second edition. Mathematical Surveys and Monographs, 155. American Mathematical Society, Providence, RI, 2009.
- [18] MUNKRES, JAMES R. *Topology: a first course*. Prentice-Hall, Inc., Englewood Cliffs, N.J., 1975.
- [19] NEWELSKI, LUDOMIR. *The diameter of a Lascar strong type*. Fund. Math. 176 (2003), no. 2, 157–170.
- [20] PELÁEZ, RODRIGO. *About the Lascar group*, Ph.D. thesis, Universitat de Barcelona. 2008.
- [21] SCHNEIDER, SAMUEL. *Simultaneous reducibility of pairs of Borel equivalence relations*, [ArXiv](#), 2013.
- [22] SILVER, JACK H. *Counting the number of equivalence classes of Borel and coanalytic equivalence relations*. Ann. Math. Logic 18 (1980), no. 1, 1–28.
- [23] SRIVASTAVA, SHASHI M. *A course on Borel sets*. Graduate Texts in Mathematics, 180. Springer-Verlag, New York, 1998.
- [24] TENT, KATRIN; ZIEGLER, MARTIN. *A course in model theory. Lecture Notes in Logic*, 40. Association for Symbolic Logic, Cambridge University Press, Cambridge, 2012.